

Explicit block-growth bounds and finite-cycle exclusions for the $3n+1$ problem

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Research review edition - independent mathematical review pending

The arguments and computational claims below have internal checks; independent mathematical review and priority confirmation are pending.

We give an explicit bound for every positive Collatz orbit, indexed by maximal odd/even blocks, with exponential base $\lambda = \log_2(3) - 1/80$. A retained elementary loss, two height strata, explicit fourth-power interpolation parameters, an unbounded dyadic argument and a restart envelope yield the bound. Published cycle estimates then give $K < 1.61m \cdot \lambda^m$ for $1024 \leq m \leq 515619$ and $K < 15.27m \cdot \lambda^m$ for $m \geq 515620$. Separately, exact finite certificates support a candidate exclusion through 100 local minima, extending the journal result through 91 and the located preprint claim through 95. We give the mathematical reduction, source-bound evidence, and reproduction instructions. Neither result proves the Collatz conjecture.

1. Statements and conventions

Use $T(n) = (3n+1)/2$ for odd positive n and $T(n) = n/2$ for even n . A block consists of a maximal odd run followed by a maximal even run. At its odd start n_t define $k_t = v_2(n_t + 1)$ and $y_t = \log_2(n_t + 1)$. Let

$$\delta = \log_2(3), \beta = \delta - 1, \varepsilon = 1/80, \\ \lambda = \delta - \varepsilon, B = 1272000, A = (\delta/\lambda)^{32}.$$

Theorem 1 (analytic bound). For every positive orbit, from every odd block start, the following estimates hold:

$$k_t \leq A \cdot y_0 \cdot \lambda^t \quad (t \geq 0), \\ y_t \leq \delta \cdot A \cdot y_0 \cdot \lambda^{t-1} \quad (t \geq 1).$$

Numerically λ is approximately 1.572462500721156 and A approximately 1.288362904142. Theorem 2 (cycle bound). For a hypothetical nontrivial positive cycle with m local minima and K odd shortcut steps,

$$K < 1.61m \cdot \lambda^m, 1024 \leq m \leq 515619, \\ K < 15.27m \cdot \lambda^m, m \geq 515620.$$

The index m counts blocks, not total cycle length. Theorem 3 (computer-assisted candidate). The completed certificates exclude $92 \leq m \leq 100$. Together with Hercher [4] this gives exclusion through 100, subject to the stated external inputs and independent review. The proof chain through 100 does not depend on any unfinished $m=101$ computation.

The comparison for Theorem 2 is the explicit δ^m upper bound in Simons-de Weger [2]. At $m=1024$ the displayed new bound is over 3,000 times smaller: approximately $3.28 \cdot 10^{204}$ instead of $9.99 \cdot 10^{207}$. Both are still very large. A smaller exponential base is an asymptotic improvement, not a convergence theorem; λ remains greater than one.

2. Exact block identities

Write $n=a \cdot 2^k - 1$, with a positive odd integer. Let $\ell \geq 1$ be the following maximal even-run length, $b_0 = 2^\ell - 1$, and n' the next odd block start. Then

$$\begin{aligned} n' &= (a \cdot 3^k - 1) / 2^\ell, \\ k' &= v_2(a \cdot 3^k + b_0) - \ell, \\ y &= k + \log_2(a). \end{aligned}$$

With $D = \ell + \beta \cdot \log_2(a) - 1 \geq 0$, elementary logarithmic estimates give

$$\begin{aligned} y' &\leq \delta \cdot y - D, \\ y'' &\leq \delta \cdot y - D + \beta \cdot k'. \end{aligned} \quad (1)$$

For example, $n'+1 < (a \cdot 3^k) / 2^\ell + 1 < 2 \cdot a \cdot 3^k / 2^\ell$ gives the first bound. The second also uses $y'' \leq y' + \beta \cdot k'$, which follows by applying the first bound to the next block and discarding $\ell' - 1 \geq 0$. Always $k \leq y$ and $y' \leq \delta \cdot y$.

3. The published interpolation criterion

We use Bugeaud (1999), Theorem 1, rational clause (6) [1]. In the rational, 2-adic setting used below, the theorem has the following form. For two rational 2-adic units α_1, α_2 and positive integers $b_1 = H, b_2 = 1$, let $\Lambda = \alpha_1^H - \alpha_2$ be nonzero. Our choices have $g=1, u=0$, field parameters $D_{\text{field}} = e=f=1$ and $t=1$, with $v_2(\alpha_1 - 1) \geq E > 1$ and $v_2(\alpha_2 - 1) \geq 2$.

Rename the theorem's integer K as M . For integers $M \geq 3, L \geq 2$ and positive integers R_1, S_1, R_2, S_2 , set

$$\begin{aligned} R &= R_1 + R_2 - 1, \quad S = S_1 + S_2 - 1, \quad N = ML, \\ \gamma &= 1/2 - N/(6RS), \quad P_M = \text{product}(i!, i=1, \dots, M-1), \\ b &= [R-1 + (S-1)H] \cdot P_M^{-2/[M(M-1)]/2}. \end{aligned}$$

The two cardinality requirements will be checked explicitly in each range. If they hold and

$$\begin{aligned} &M(L-1) \cdot E \cdot \ln 2 - 3 \ln N - (M-1) \ln b \\ &\quad - \gamma \cdot L \cdot R \cdot h(\alpha_1) - \gamma \cdot L \cdot S \cdot h(\alpha_2) > 0, \end{aligned}$$

then $v_2(\Lambda) \leq E(ML-1/2) < EML$. For a reduced rational p/q , its logarithmic height is $h(p/q) = \ln(\max(|p|, |q|))$. Negative rational arguments are allowed by the theorem. Once the stated cardinalities hold, this criterion requires no multiplicative-independence assumption.

4. A factorial estimate used in both ranges

For every integer $M \geq 800$,

$$P_M^{-2/[M(M-1)]} < (23/5)/(M-1). \quad (2)$$

Indeed $\ln(i!) \geq i \ln(i) - i + 1$ by integrating $\ln x$. Since $x \ln x$ is increasing for $x \geq 1$, a second integral comparison gives

$$\begin{aligned} \ln P_M &\geq ((M-1)^2/2) \ln(M-1) - (M-1)^2/4 + 1/4 \\ &\quad - M(M-1)/2 + M-1. \end{aligned}$$

Multiplying by $-2/[M(M-1)]$ gives a quantity at most $-\ln(M-1) + 3/2 + \ln(M-1)/M$. For $M=800u, u \geq 1$, $\ln(M-1) < \ln 800 + \ln u < 6.7 + u - 1 < 7.2u$, using $\ln u \leq u - 1$. Consequently $\ln(M-1)/M < .009$. Finally $1.509 < \ln(23/5)$,

proving (2). We may also use the weaker factor $5/(M-1)$.

The numerical bounds used throughout are

$$\begin{aligned} .693 < \ln 2 < .6932, \ln 3 < 1.099, \ln 264 < 5.576, \\ \ln(7/4) < .56, \ln 800 < 6.7, \ln 11 < 2.4. \end{aligned}$$

The additional bounds $\ln 90 < 4.5$ and $\ln 150 < 5.011$ follow from $\ln 90 = \ln 2 + 2\ln 3 + \ln 5$ and $\ln 150 = \ln 2 + \ln 3 + 2\ln 5$ using exact series enclosures. Every numerical logarithm bound used here has a directed rational check in the accompanying audit.

5. Retaining the elementary loss

Write $n = a \cdot 2^k - 1$ with a odd, and let ℓ be the following even-run length. Set

$$D = \ell + \beta \cdot \log_2(a) - 1 \geq 0.$$

The exact elementary block inequalities used before are

$$\begin{aligned} y' &\leq \delta \cdot y - D, \\ y'' &\leq \delta \cdot y - D + \beta \cdot k' \\ &= \delta^2 \cdot y - D - \beta \cdot (\delta \cdot y - k'). \end{aligned}$$

The previous local proof bounded the last loss using only the β term. Here we keep D when it is large; when D is small the logarithmic height in the interpolation theorem becomes smaller. These are two exhaustive alternatives, not assumptions about typical trajectories.

Fix J divisible by 100, $J \geq 10000$, and $y \geq 79.5J$. If $D \geq J$, the one-block alternative $y' \leq \delta \cdot y - J$ already suffices. Otherwise split into

$$\text{I: } 0 \leq D \leq .9J, \text{ II: } .9J < D < J.$$

In either stratum write $a_- = 0, a_+ = .9$ for I and $a_- = .9, a_+ = 1$ for II. Then $\ell < a_+ J + 1$ and $\log_2(a) < (12/7)(a_+ J + 1)$, using $\beta > 7/12$. The resulting strict upper bound is safe also at the closed boundary $D = .9J$: $\ell \leq D + 1$ and $\beta \cdot \log_2(a) = D - \ell + 1 \leq D$. In particular $k > 77J$.

6. Fourth-power normalization and hypotheses

Choose r_0 in $\{0, 1, 2, 3\}$ so that $k + r_0$ is divisible by 4, and $H = (k + r_0)/4$. Apply Bugeaud (1999), Theorem 1, rational clause (6), with the following choices, with

$$\begin{aligned} \alpha_1 &= 81, \alpha_2 = -3^r_0 \cdot (2^\ell - 1)/a, \\ \Lambda &= 81^H - \alpha_2, \\ p &= 2, g = 1, E = 4, b_1 = H, b_2 = 1, u = 0, \\ D_{\text{field}} &= e = f = t = 1. \end{aligned}$$

If $v_2(\Lambda) \leq 3$, all bounds below are immediate. Otherwise α_2 is 1 modulo 16 and both required valuations are at least 4. Signed rational arguments are permitted by the cited theorem. The result is nonzero because $\alpha_1^H > 0 > \alpha_2$. Reduction of the rational α_2 cannot increase its height.

The logarithmic heights satisfy

$$\begin{aligned}h(\alpha_1) &= 4\ln 3, \\h(\alpha_2) &< U \cdot J, \\U &= (12/7)(a_+ + 1/10000) \cdot .69314719.\end{aligned}$$

For the numerator, its binary logarithm is less than $\ell + 3\delta$, which is less than $a_+ J + 1 + 3\delta < (12/7)(a_+ J + 1)$; the last inequality is uniform for $J \geq 10000$ and $a_+ \geq 9$. The denominator follows from the preceding bound for a .

7. Exact parameters for the two loss strata

For $M \geq 60000$, the already proved integral estimate gives

$$\begin{aligned}& -2\ln(\text{product}(i!, 1 \leq i < M)) / (M(M-1)) \\& \leq -\ln(M-1) + 3/2 + \ln(M-1)/M.\end{aligned}$$

Write $M = 60000u$, $u \geq 1$. Since $\ln 60000 < 12$ and $\ln u \leq u-1$, $\ln(M-1) < 12u$, so the last quotient is less than .0002. Exact rational logarithm enclosures prove $1.5002 < \ln 4.483$. Therefore the factorial factor in the interpolation criterion is strictly less than $4.483/(M-1)$.

Set $M = xJ, L = 5, R_1 = 5, S_1 = 1, R_2 = rJ, S_2 = S$. Thus $R = rJ + 4, N = 5xJ$ and $\gamma = 1/2 - N/(6RS)$. Use these rows in order within each stratum:

Stratum	Lower height/J for the loss	Upper k/J	x	r	S	Valuation bound/J
I	79.5	102.41	6.02	3.01	10	120.4
I	102.41	256	7.84	3.92	10	156.8
II	79.5	79.95	6.10	3.39	9	122.0
II	79.95	80.92	6.13	3.41	9	122.6
II	80.92	86.03	6.21	3.45	9	124.2
II	86.03	109.43	6.62	3.68	9	132.4
II	109.43	256	8.47	4.24	10	169.4

The first row of each stratum uses the assumed $y \geq 79.5J$. Subsequent rows apply after k exceeds the preceding endpoint, and use $y \geq k$. All parameters are integers because 100 divides J ; $M \geq 60200$. Also $R_2 < H$ because $k > 77J$ and $r \leq 4.24$. Distinct powers of 81 prove the first cardinality condition; injectivity of $z + Hw$ on $0 \leq z < R_2, 0 \leq w < S$ proves the second, because $rS \geq 5x$ and hence $R_2 \cdot S \geq ML > (M-1)L$. No multiplicative independence is needed.

For upper endpoint T , the normalized interpolation parameter obeys

$$b < 4.483 \cdot [rJ + 3 + (S-1)(TJ + 3)/4] / [2(xJ - 1)].$$

The right side decreases with J . Moreover

$$\begin{aligned}\gamma \cdot R/J &\leq r/2 + .0002 - 5x/(6S), \\ \gamma \cdot S &\leq S/2 - 5x/[6(r + .0004)], \\ 3\ln N/J &< .0039.\end{aligned}$$

For the last inequality use $J=10000v$, $N \leq 423500v$, $\ln 423500 < 13$ and $\ln v \leq v-1$. The exact rational auditor encloses $\ln b$ at $J=10000$ upward to five decimal places. With that upper bound L_b , it checks the positive margin

$$\begin{aligned} & 16x \cdot .69314718 - x \cdot L_b - .0039 \\ & -5 \cdot \{[r/2 + .0002 - 5x/(6S)] \cdot 4 \cdot 1.09861229 \\ & + [S/2 - 5x/(6(r+.0004))] \cdot U\} > 0. \end{aligned}$$

The logarithmic constants are all checked by directed rational series. This is precisely a sufficient lower bound for the published interpolation criterion divided by J , uniformly for every admissible J . It follows that $v_2(\Lambda) < 4ML = 20xJ$, and therefore $k' < 20xJ$. The companion exact audit records every rational margin.

8. Local loss in every range

For each row let h be its lower height/ J . The auditor checks exactly

$$\begin{aligned} g &= 1.5849625 \cdot h - 20x > 1, \\ a_- + .5849625 \cdot g &> 16/5. \end{aligned}$$

Consequently $k' \leq \delta \cdot y - J$ and

$$\begin{aligned} y'' &\leq \delta^2 \cdot y - D - \beta \cdot (\delta \cdot y - k') \\ &\leq \delta^2 \cdot y - (16/5)J. \end{aligned}$$

The term a_- is retained in stratum II. For $k/J > 256$, the following unbounded dyadic argument gives $k' < (38/25)k$. Its hypotheses $J \geq 100$ and $D < J$ hold here. The weaker exact bounds $\delta > 19/12, \beta > 7/12$ already imply $\beta \cdot (\delta - 38/25) \cdot 256 > 16/5$ and $\delta - 38/25 > 1/256$. Thus the same local alternative holds beyond the finite table. This covers every k and D .

9. The unbounded dyadic argument

Assume $D < J$ and $k/J > 256$. Put $r = k \bmod 2$, $H = (k+r)/2$, $\alpha_1 = 9$ and $\alpha_2 = -3^r \cdot b_0/a$. Then $\Lambda = 3^r \cdot (a \cdot 3^k + b_0)/a > 0$. If its valuation is at most 2 the desired bound is immediate. Otherwise α_2 is congruent to 1 modulo 8. Thus the rational criterion applies with $E=3$ and the same field and exponent parameters as before. Here $h(\alpha_1) = 2 \ln 3$ and $h(\alpha_2) < U_0 \cdot J$, where $U_0 = 525099/437500$ follows from $(12/7) \cdot (101/100) \cdot .6932$ and $J \geq 100$. Below, $\gamma > 0$.

Let q be the least integer with $k/J \leq 2^q$. Then $q \geq 9$ and $k/J > 2^{q-1}$. Choose

$$\begin{aligned} M &= (q+1)J, L = q+2, R_1 = L, S_1 = 1, \\ R_2 &= (q-1)J, S_2 = q+5. \end{aligned}$$

Thus $R = (q-1)J + q+1, S = q+5, N = (q+1)(q+2)J$. Here $M \geq 1000$. The inequality $q-1 < 2^{q-2}$ gives $R_2 < H$. Also $R_2 \cdot S_2 = (q^2 + 4q - 5)J \geq ML > (M-1)L$, because $q \geq 9$. Again $RS > N$ and $1/3 < \gamma < 1/2$.

The weaker factorial factor 5, and $H \leq (2^q J + 1)/2$, give

$$\begin{aligned} b &< 5 \cdot \{[2(q-1) + (q+4)2^q]J + 3q+4\} / [4((q+1)J-1)] \\ &\leq (7/4) \cdot 2^q. \end{aligned}$$

For the last step it suffices that $J[(2q-13)2^q - 10q+10] \geq 7 \cdot 2^q + 15q+20$. At $J \geq 100$ this follows from $(200q-1307)2^q \geq 1015q-980$. The latter holds for $q \geq 9$ already on replacing 2^q by 512. Hence $\ln b < .6932q + .56$.

The exact γ formulas imply

$$\begin{aligned}\gamma \cdot R/J &= q/3 - 1/6 - 2/(q+5) + (q+1)/(2J) \\ &\leq (203/600)q - 97/600, \\ \gamma \cdot S &\leq [1/2 - 350/2127]q + 5/2 - 1400/2127.\end{aligned}$$

For the second bound use $(q+1)/[(q-1)J] \leq 9/700$ and $(q+1)(q+2)/(q-1) = q+4+6/(q-1)$. Finally $3 \ln N/J < q/25$: $\ln J < J/20$, $\ln(q+2) < q/3$, and $\ln(q+1) < q/3$ imply $3 \ln N/J < .15 + .02q < .04q$ at $q \geq 9$. The logarithm inequalities follow by writing $J = 100u$ or $q+2 = 11u$ and using $\ln u \leq u-1$, $\ln 100 < 4.7$ and $\ln 11 < 2.4$.

The interpolation margin, divided by J , is therefore greater than

$$\begin{aligned}P(q) &= (q+1)[(q+1) \cdot 2.079 - .6932q - .56] - q/25 \\ &\quad - (q+2)\{[(203/600)q - 97/600] \cdot 2.198 \\ &\quad + [(1/2 - 350/2127)q + 5/2 - 1400/2127] \cdot U_0\}.\end{aligned}$$

Its exact coefficients, from quadratic to constant, are

$$\begin{aligned}1783170953/7444500000, \\ -454813329/354500000, \\ -1631430293/744450000.\end{aligned}$$

The leading coefficient is positive, $P(9) = 3011630363/531750000 > 5.66$, and $P'(9) = 1503066483/496300000 > 3.02$. Expansion at 9 proves $P(q) > 0$ for every real $q \geq 9$. The theorem yields $k' < 3J(q+1)(q+2)$. At $q=9$, $3(q+1)(q+2) = 330 < (38/25) \cdot 256$. The ratio of consecutive left sides is $(q+3)/(q+1) \leq 2$, so induction proves

$$3(q+1)(q+2) < (38/25) \cdot 2^{q-1} < (38/25) \cdot (k/J).$$

The first cardinality is L because $9^{2z}, 0 \leq z < L$, are distinct. The second follows from $R_2 \leq H$, which makes $z + Hw$ injective on the stated rectangle. Thus all hypotheses are checked, and this proves $k' < (38/25)k$ throughout $k/J > 256$.

10. Global restart induction

Set $C = 79.5$. For $y \geq B$ choose $J = 100 \cdot \text{floor}(y/(100C))$. Then J is a multiple of 100, and the exact conditions

$$\begin{aligned}C \cdot 10000 &= 795000 \leq B, \\ 100C/(1 - C/80) &= 1272000 = B\end{aligned}$$

give $J \geq 10000$, $y \geq CJ$ and

$$J > y/C - 100 \geq \varepsilon \cdot y.$$

The preceding alternatives therefore bound either the next height by $\lambda \cdot y$, or both the intermediate odd-run length by $\lambda \cdot y$ and the two-block height by $\lambda^2 \cdot y$. The latter follows from $2\delta \cdot \varepsilon - \varepsilon^2 < (16/5)\varepsilon$.

Define $G(y) = \delta \cdot y$ for $y \leq B$, and $G(y) = \max(\lambda \cdot y, y + \beta \cdot B)$ for $y \geq B$. It is continuous and increasing, and $\lambda \cdot y \leq G(y) \leq \delta \cdot y$. Below B use the elementary one-block inequality. Above B use the proved alternative and monotonicity. Induction along the selected restart times bounds every restart endpoint by the corresponding virtual iterate $G^t(y_0)$. At the single intermediate index of a two-block restart, the checked run bound gives $k_t \leq G^t(y_0)$, while $y' \leq \delta \cdot y$ gives $y_t \leq \delta \cdot G^{t-1}(y_0)$. At restart endpoints the latter follows from $G(y) \leq \delta \cdot y$. Thus every block index is covered, not just restart times.

For $y \geq \beta \cdot B / (\lambda - 1)$, $G(y) = \lambda \cdot y$. Directed rational checks give $\lambda > 1.57$ and $1.03(\lambda - 1) > \beta$. The integer inequality

$$157^{32} > 103 \cdot 1272000 \cdot 100^{31}$$

therefore puts every virtual orbit from $y_0 \geq 1$ in the linear region by step 32. Before that time use $G \leq \delta \cdot y$, and afterwards $G = \lambda \cdot y$. Consequently $G^t(y_0) \leq A \cdot y_0 \cdot \lambda^t$, including $t < 32$. This proves Theorem 1. The virtual envelope reaching its linear regime does not assert that an actual orbit grows or reaches any particular height.

11. Cycle consequences and comparison

The same extension also satisfies, for every $y \geq 1$,

$$G(y) \leq \lambda \cdot y + \varepsilon \cdot B.$$

For $y \leq B$ this is $\varepsilon \cdot y \leq \varepsilon \cdot B$. For $y \geq B$ the linear term is immediate; for the other term subtracting the right side leaves $(\lambda - 1)(B - y) \leq 0$. Set $c = \varepsilon \cdot B / (\lambda - 1)$. Induction gives

$$G^t(y_0) \leq (y_0 + c) \cdot \lambda^t - c. \quad (3)$$

For a hypothetical nontrivial cycle, rotate to its minimum. Simons-de Weger [2], Theorem3(d), gives $K < 16m \cdot \delta^m$ for $m \geq 91$ after coarsening its three applicable coefficients. Corollary13 gives $n_{\min} < m \cdot \exp(13.3 \cdot (.46057 + \ln K))$. Consequently, with $a_0 = \log_2(\delta)$,

$$y_0 < 1 + 13.3 \cdot .46057 / \ln 2 + 53.2 + 14.3 \log_2(m) + 13.3 a_0 \cdot m.$$

For $m \geq 1024$ use $\ln 2 > .69$, $a_0 < 2/3$, and $\log_2(m) \leq m/64$. The constant term is below 64, and $13.3 \cdot (2/3) + 14.3/64 + 64/1024 < 10$. The logarithm inequality holds at 1024 and its difference increases thereafter. The bound $a_0 < 2/3$ follows from $\delta < 317/200$ and $(317/200)^3 < 4$. Thus $y_0 < 10m$. Exact rational coefficient bounds now imply

$$K < 22.51m \cdot \lambda^m, \quad m \geq 1024,$$

$$K < 1.61m \cdot \lambda^m, \quad 1024 \leq m \leq 515619.$$

For the second bound, Theorem3(d) gives $n_{\min} < 51825000 \cdot m^2 \cdot \delta^m$ when $91 \leq m \leq 515619$. Hence

$$y_0 < Y(m) = 1 + \log_2(51825000) + 2 \log_2(m) + a_0 \cdot m.$$

Since $51825000 < 2^{26}$ and $\log_2(m)/m$ decreases for $m \geq 1024$, $Y(m)/m < 2/3 + 47/1024$. The exact audit checks $[A/(\lambda - 1)] \cdot [2/3 + 47/1024] < 1.61$. Summing the block estimates therefore gives the stated middle-range coefficient.

There is a stronger large-range coefficient. Substitute the first bound into Simons-de Weger, Corollary13. With

$$Q = 1 + 13.3 \cdot .46057 / \ln 2 + 13.3 \log_2(22.51),$$

we obtain $y_0 < Q + 14.3 \log_2(m) + 13.3m \cdot \log_2(\lambda)$. Summing (3) around the cycle consequently gives

$$\begin{aligned} K &< [(y_0 + c)/(\lambda - 1)] \cdot \lambda^m, \\ K &< 15.27m \cdot \lambda^m, \quad m \geq 515620. \quad (4) \end{aligned}$$

For (4), the coefficient upper bound

$$[13.3 \log_2(\lambda) + (Q + 14.3 \log_2(m) + c)/m] / (\lambda - 1)$$

decreases for $m \geq 515620$, since both $1/m$ and $\log(m)/m$ decrease there. At $m = 515620$ an exact upward rational calculation puts it below 15.27. This uses an already established bound as input to Corollary13, so

the argument is not circular. This proves Theorem 2. The affine estimate improves the coefficient; the fourth-power normalization and retained loss improve the base.

12. Sharp majorization for cyclic growth constraints

Let b be real, let $m \geq 1$, and let $\Phi: [b, \infty) \rightarrow [b, \infty)$ be continuous and strictly increasing, with $\Phi(x) \geq x$. Define its truncated inverse Ψ by

$$\begin{aligned}\Psi(t) &= b, \quad b \leq t \leq \Phi(b), \\ \Psi(t) &= \Phi^{-1}(t), \quad t > \Phi(b).\end{aligned}$$

Then Ψ is continuous, nondecreasing, and $\Psi(t) \leq t$. For any $S \geq m \cdot b$ there is a unique $t \geq b$ such that

$$w_j = \Psi^{m-j}(t), \quad j=1, \dots, m, \quad \sum_j w_j = S.$$

Existence and uniqueness follow because this sum is continuous and strictly increasing in t (its last term is t), starts at $m \cdot b$, and tends to infinity.

Theorem 4. The increasing vector w majorizes every cyclic vector x with $x_i \geq b$, $x_{i+1} \leq \Phi(x_i)$, and $\sum_i x_i = S$. Consequently, for every convex f ,

$$\sum_i f(x_i) \leq \sum_j f(w_j).$$

The bound is attained by w itself arranged in increasing cyclic order.

Proof. Sort x into $u_1 \leq \dots \leq u_m$. For every $j < m$, an edge of the original cycle goes from the set of j smallest vertices to its complement. Its starting value is at most u_j and its ending value is at least u_{j+1} . Monotonicity gives

$$u_{j+1} \leq \Phi(u_j), \quad \text{equivalently } u_j \geq \Psi(u_{j+1}).$$

Suppose an increasing prefix through j had sum smaller than the corresponding w -prefix. If $w_j = b$ this is impossible because all $u_i \geq b$. Otherwise $u_j < w_j$: indeed $u_j \geq w_j$ would imply

$$u_i \geq \Psi^{j-i}(u_j) \geq \Psi^{j-i}(w_j) = w_i$$

for every $i \leq j$, contradicting the smaller prefix sum. Since $w_j > b$, all subsequent inverse steps are untruncated, so $w_{j+r} = \Phi^r(w_j)$. Strict monotonicity now gives $u_{j+r} < w_{j+r}$ for every $r \geq 1$. The smaller prefix and smaller remaining suffix contradict equality of total sums. Thus every increasing prefix of u is at least the corresponding prefix of w , which is exactly the majorization assertion. Finally $w_{j+1} \leq \Phi(w_j)$, including at the floor transition; the wraparound inequality holds because $w_1 \leq w_m \leq \Phi(w_m)$. This proves feasibility and sharpness. End of proof.

The linear geometric-ramp lemma is the special case $\Phi(x) = d \cdot x$ after shifting the floor. The proof does not require Φ to be concave or differentiable.

13. A subordinate envelope for periodic run lengths

Suppose a periodic sequence of actual heights $y_i \geq b$ and positive odd-run lengths k_i satisfies

$$k_{i+t} \leq \Phi^t(y_i) \quad \text{for every } t \geq 0.$$

Write $v_i = \max(b, k_i)$, and set

$$z_i = \max_{t \geq 0} \Psi^t(v_{i+t}).$$

It is enough to take $0 \leq t < m$: periodicity and $\Psi^m(x) \leq x$ make every later term no larger than its predecessor m places earlier. The future-run bounds and $y_i \geq b$ give $z_i \leq y_i$. Also $z_i \geq v_i$ and

$$z_i = \max(v_i, \Psi(z_{i+1})),$$

so $z_{i+1} \leq \Phi(z_i)$. Thus $\sum_i z_i \geq \sum_i k_i = K$.

If $K \geq m \cdot b$ and $\sum_i z_i > K$, cap every z_i at a common $T \geq b$ until the sum is K . This preserves the growth constraint for any Φ as above: if $z_i \leq T$ the previous edge inequality remains sufficient; if $z_i > T$, the capped successor is at most $T \leq \Phi(T)$. A suitable T exists by continuity of $\sum_i \min(z_i, T)$. This avoids a concavity assumption that a straight-line interpolation would require.

For any decreasing convex f , capping only increases the objective. Applying the sharp theorem gives an upper bound for $\sum_i f(y_i)$ from the extremal vector of total K . A proved lower bound $K_0 \leq K$ can be used in exactly the same way by capping to total K_0 , provided $K_0 \geq m \cdot b$. If $K_0 < m \cdot b$, the direct bound $m \cdot f(b)$ applies.

For Collatz, use $f(y) = 1/(2^y - 1)$. This converts a certified nonlinear future-run bound into an exact profile bound for the cycle logarithmic identity.

14. Certified finite-cycle application

We also give a computer-assisted candidate exclusion of nontrivial positive cycles with 92 through 100 local minima. Together with Hercher's exclusion through 91 [4], including its published corrigendum, this would exclude all nontrivial positive cycles with at most 100 local minima. Here m is the number of maximal odd/even blocks, not the number of integers in a cycle. The strongest analytic theorem above is a separate result: the completed finite certificates use earlier, already frozen specializations with $\epsilon = 1/160$ or $1/93$. No unfinished stronger-map experiment is an input to these claims.

The external computational input is Barina's published verification below $X = 2^{71}$ [3]; X itself is a power of two and converges. Consequently a hypothetical nontrivial cycle has every member greater than X . We have not repeated that full convergence verification. Wang's 2026 preprint [5] states an exclusion through 95 and supplies parity-prefix code. We inspected the manuscript and the latest downloadable source, but do not use its claimed exclusion as an input: our exact profiles independently cover 92 through 95.

The finite restart map

Define, on $y \geq 71$,

$$\Phi(y) = \max(\delta \cdot y - 40.99, \min(\delta \cdot y - 37.99, y + 71 \cdot (\delta - 1) - 37.99)).$$

This continuous increasing map has slopes at least one and $\Phi(y) \geq y$. A sufficient restart witness at a block start is a finite $r \geq 1$ with

$$k_j \leq \Phi^j(y_0) \text{ for } 0 \leq j < r, y_r \leq \Phi^r(y_0).$$

Monotonicity and strong induction then give $k_t \leq \Phi^t(y_0)$ for every t . Every intermediate run must be checked. Alternatively, verified entry into the convergence basin eliminates the starting value from any nontrivial cycle.

The finite reduction uses the elementary one- and two-block inequalities, Bugeaud (2002), Theorem 2 [8], and an exhaustive modular sweep. It confines exceptions to $k < 200$, $\ell \leq 40$, and odd parts below 2^{69} . The large-exponent estimate is $v_2(a \cdot 3^k + 2^\ell - 1) < 1700 \cdot (\ln k + 2)^2 + 3 < k$ for $k \geq 2^{19}$; the last comparison extends from its cutoff by monotonicity. For the remaining range, the modular sweep verifies $40 \cdot (2^{19} - 1) = 20,971,480$ pairs and the bound on the next odd-run length $s \leq 100$. Exact sufficient inequalities remove the range $k \geq 200$. The

surviving input consists of 44,497 residue classes containing exactly 144,429,913,922,369,056,364 seeds.

Two complete algorithms discharge that same input. The first branches using inverse residues and exact affine formulas; the second bisects arithmetic progressions by parity. They use 1,648,100,522 and 1,253,664,985 nodes, respectively. Every class finishes, coverage totals agree, and the maximum restart depth is five. Their different partition rules give different node counts. Neither count is presented as an independent mathematical theorem.

For a progression of odd parts a , a later minimum has the form $n_j = (P \cdot a + Q)/2^S$ with $P > 0$ and $Q + 2^S \geq 0$. The restart difference $\Phi^j(k_{\text{root}} + \log_2(a)) - \log_2(n_j + 1)$ is nondecreasing: after multiplying its derivative by $a \cdot \ln 2$, it is at least

$$1 - P \cdot a / (P \cdot a + Q + 2^S) \geq 0.$$

Continuity covers the breakpoints of Φ . Thus a successful least-endpoint test certifies the entire progression. Whole-family basin tests instead use the greatest endpoint. Integer logarithm bounds and a rational mantissa grid give directed comparisons. Python independently checks all 2,032,128 grid thresholds and 112 selected progression cases, including difficult cases. It does not repeat all 44,497 classes. The full two-algorithm proof is native integer code, with its sources, inputs, counters and hashes retained.

Joining the analytic tail

For each of $(\varepsilon, B) = (1/160, 16384)$ and $(1/93, 32768)$, define

$$F(y) = \min(\Phi(y), (\delta - \varepsilon) \cdot y + \varepsilon \cdot B - 40.99).$$

The earlier analytic specializations establish a one- or two-block restart above B . Their proofs are included in the accompanying sources. The two formulas meet at B , and F is continuous, increasing, at least the identity, with slopes at least one. Merely taking a minimum of restart maps would not prove a restart property; the following extra check is essential.

Above B , the analytic restart transfers because $\varepsilon \cdot B - 40.99 > 0$ and $F(y) \geq y$. Below B , all finite witnesses stay below the joining point: the maximum initial upper height is 117 and $\delta^5 \cdot 117 < B$. Nonexceptional small- k witnesses satisfy $\delta^2 \cdot 300 < B$. In the middle range $k \geq 200$, the old sweep gives $s \leq 100$, and the global lower bound $F(y) \geq (\delta - \varepsilon) \cdot y - 40.99$ gives a two-block witness from the exact positive margin

$$\begin{aligned} & ((\delta - \varepsilon)^2 - \delta) \cdot 200 \\ & - (\delta - \varepsilon + 1) \cdot 40.99 - (\delta - 1) \cdot 100 > 0. \end{aligned}$$

The companion intermediate-run check is $(\delta - \varepsilon) \cdot 200 - 40.99 > 100$. These checks handle a virtual step crossing B . The source-bound graft audits verify these inequalities and the actual finite root heights and depths. Thus each F is valid for every basin-avoiding orbit, in particular every hypothetical nontrivial cycle.

Rational approximation and the finite window

Let H be the total number of binary divisions in a hypothetical shortcut cycle and K its number of odd steps. The cycle product identity and the geometric bound for reciprocals along each odd run give

$$0 < H/K - \delta < (1/(K \cdot \ln 2)) \cdot \sum_i 1/n_i,$$

where n_i are the m local minima. Apply the growth-constrained majorization theorem proved above to F and $f(y) = 1/(2^y - 1)$. An established lower bound $K_0 \leq K$ gives an explicit upper bound for the reciprocal sum. Directed rational intervals for δ and $\ln 2$ then give an open interval containing H/K . Exact Stern-Brocot bracketing finds its minimum possible denominator and therefore a new lower bound on K . The saved determinant-one neighbors certify that step independently. Iterate until the lower bound exceeds the

applicable published upper bound $1.4784 \cdot m \cdot \delta^m$ [2].

For $m=96,97,98$ the completed profiles close directly at X . The cases 99 and 100 first require excluding least minima in $[X,11X]$ and $[X,15X]$, respectively. Necessary prefix targets come from the remaining run capacity: with $A_r(y) = \sum_{j=0}^{r-1} F^j(y)$, a prefix of i blocks obeys

$$A_{m-i}(y_i) \geq K_0 - \text{ceil}(A_i(B_0)),$$

where B_0 is a proved upper bound on the starting height in that window. Upward rational evaluation of A and a positive-series lower bound for 2^h certify necessary integer targets for each later minimum.

The exact prefix search represents $n_i = (P \cdot n_0 + Q)/2^S$. Specifying an odd run and its following even run adds congruences modulo powers of two; intersecting them with the original interval produces disjoint residue classes. A branch closes only by an empty class, descent below the assumed global minimum, or violation of a necessary target. Singleton branches are iterated exactly. A resource or depth limit is unresolved, never a successful exclusion. Independent partition audits reconstruct coverage from the source configuration and verify that the complete job list covers the window.

m	Minimum factor after search	Prefix nodes	Completed verification
96	1	No additional window	Exact forward and inverse profiles
97	1	No additional window	Exact forward and inverse profiles
98	1	No additional window	Exact forward and inverse profiles
99	11	2,758,731,446	Full original-integer repeat and all audits
100	15	14,528,184,145	Complete partition, journal and fallback audits

The $m=99$ repeat uses a different initial partition; after accounting for the partition boundaries its seven substantive counters agree exactly. For $m=100$, the single job that invokes arbitrary-precision fallback was also repeated with the original arithmetic, using 5,063,641 nodes. A whole-search original-arithmetic repeat remains in progress and is not counted as completed evidence. Separately completed, larger $m=100$ searches provide another route to the same exclusion; the compact package uses the 15X route in the table.

For $m=99$, the final exact profile lower bound is

$$K \geq 7797285910967231555816016771572875912,$$

whereas the rounded published upper ceiling is 9274858856993487264329. For $m=100$ the corresponding quantities are

$$K \geq 3876482907693838530030634129025455,$$

$$K < 14848791401834506924263.$$

Each is a contradiction. The end-to-end checks bind the map certificates, profile reconstruction, target derivation, complete searches, partition audits and fallback validation. These are internally checked computational proof claims, not external peer review.

15. Verification, prior work and significance

There are two proposed advances. The analytic result supplies an explicit all-orbit block-growth base below δ and an asymptotically smaller cycle upper bound than the δ^m bounds in Simons-de Weger [2]. The finite result extends the completed candidate exclusions beyond Hercher's journal result through 91 [4] and Wang's preprint claim through 95 [5]. The methods explicitly build on their work, Bugeaud's p-adic estimates [1,8], and Barina's computation [3]. Standard affine parity coding and prefix pruning are not inventions of this manuscript.

Luca [6] proves effective exponential bounds relating cycle complexity and height. His run parameter counts blocks with $k_i \geq 2$ and can be smaller than our m ; we do not identify the parameters. His displayed height recurrence uses a factor at least δ , and does not state the uniform all-orbit bound with the explicit smaller base given here. Normalized-exponent refinements are established transcendence-theory tools, reviewed by Bugeaud [7]. The specific specialization and finite cases still need a complete priority check.

The analytic proof consists of universal inequalities, not extrapolation from computation. Exact rational auditors check its constants. Selected algebraic, congruence, restart and majorization lemmas have also been accepted by a pinned Lean/mathlib environment without proof placeholders. This is partial formal verification: the published transcendence theorem, complete specialization, imported cycle results and exhaustive native programs are not all formalized in Lean. Independently written checks still share mathematical assumptions and the same machine, so correlated mistakes remain possible.

The finite review archive was extracted into a fresh directory. Four end-to-end exclusion audits and two analytic audits reproduced their saved receipts byte for byte. The archive contains sources, exact inputs, journals, proof notes, hashes and rerun commands; its default audit does not silently claim to rerun every billion-node search. Longer reruns are documented. The strongest new analytic specialization has a separate exact audit and is included in the final companion package.

If independent review confirms both correctness and priority, these results would be a quantitative contribution to the study of Collatz cycles and block growth. They would not be a proof of convergence or a structural classification of every possible cycle. The new base remains greater than one, so the theorem still permits unbounded growth. Excluding any fixed number of local minima still leaves infinitely many possible cycle sizes. The work narrows rigorous bounds and supplies reusable certificates; it does not establish that the main conjecture is close to resolution.

16. Reproducibility and remaining review

The companion package includes the complete mathematical source, standalone exact auditors, native search and reference implementations, machine-readable certificates, selected Lean proofs and a pinned toolchain description. Saved hashes bind completed evidence to the source that produced it. Failed and unfinished experiments remain in the research notebook with their status; they are excluded from the claims above. No external AI model was consulted.

An external reviewer should first check the rational clause of Bugeaud's 1999 theorem, the height conventions, the cardinality conditions, the unbounded dyadic range, the restart induction, and the analytic-to-finite joining argument. The author-hosted 1999 version was inspected directly; a comparison with the final journal text remains desirable. For the finite argument the priority items are exhaustive residue coverage, directed height rounding, overflow/fallback handling and global-minimum window scope. A full original-arithmetic $m=100$ rerun and independent-machine reproduction would add confidence. None of those future checks is reported as already done.

Authorship and AI disclosure

Luke Baer and Amy (AI research system), 29 September 2026.

Luke Baer originated and directed this research project, supplied its resources, and set the requirements for substantive results, saved evidence and reproducible verification. Amy, operating through OpenAI Codex, developed the mathematical arguments, wrote the implementations and manuscript, and performed the internal checks described here. Amy is not a human researcher. The joint research credit in this review edition is not a representation that an AI system is eligible for journal authorship. Any submission must comply with the destination's human-authorship and AI-disclosure rules and receive accountable human review. No independent mathematical review has yet been obtained.

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- [8] Y. Bugeaud, Linear Forms in two m-adic Logarithms and Applications to Diophantine Problems, Compositio Mathematica 132 (2002), 137-158, Theorem 2. Bugeaud 2002: DOI 10.1023/A:1015825809661 Only Theorem 2 supplies the finite reduction. We do not silently change the height conventions in the differently printed general Theorem 1.